

Understanding Climate Change

Chapter 1: Introduction to Climate Change

Climate change refers to significant, long-term changes in the global climate. The term "global climate" encompasses the planet's overall weather patterns, including temperature, precipitation, and wind patterns, over an extended period. Over the past century, human activities, particularly the burning of fossil fuels and deforestation, have significantly contributed to climate change.

Historical Context

The Earth's climate has changed throughout history. Over the past 650,000 years, there have been seven cycles of glacial advance and retreat, with the abrupt end of the last ice age about 11,700 years ago marking the beginning of the modern climate era and human civilization. Most of these climate changes are attributed to very small variations in Earth's orbit that change the amount of solar energy our planet receives. During the Holocene epoch, which began at the end of the last ice age, human societies flourished, but the industrial era has seen unprecedented changes.

Modern Observations

Modern scientific observations indicate a rapid increase in global temperatures, sea levels, and extreme weather events. The Intergovernmental Panel on Climate Change (IPCC) has documented these changes extensively. Ice core samples, tree rings, and ocean sediments provide a historical record that scientists use to understand past climate conditions and predict future trends. The evidence overwhelmingly shows that recent changes are primarily driven by human activities, particularly the emission of greenhouse gases.

Chapter 2: Causes of Climate Change

Greenhouse Gases

The primary cause of recent climate change is the increase in greenhouse gases in the atmosphere. Greenhouse gases, such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O), trap heat from the sun, creating a "greenhouse effect." This effect is essential for life on Earth, as it keeps the planet warm enough to support life. However, human activities have intensified this natural process, leading to a warmer climate.

Fossil Fuels

Burning fossil fuels for energy releases large amounts of CO₂. This includes coal, oil, and natural gas used for electricity, heating, and transportation. The industrial revolution marked the beginning of a significant increase in fossil fuel consumption, which continues to rise today.

Coal